

Absolutely. For **Day 2**, I would continue directly from Day 1 so students understand that **OSI is a reference model**, while **TCP/IP is the practical networking model used on real networks and the Internet**.

Day 2: TCP/IP Suite

1. What is the TCP/IP Suite?

TCP/IP stands for:

Transmission Control Protocol / Internet Protocol

TCP/IP is a **set of networking protocols** used for communication between computers and devices over a network.

The Internet mainly works using TCP/IP protocols.

Simple Example

When you open:

<https://www.google.com>

your computer needs to:

1. Find Google's IP address.
2. Send data to the destination.
3. Make sure the data reaches the correct application.
4. Receive the response.
5. Display the webpage.

TCP/IP protocols handle these communication processes.

2. Why Do We Need TCP/IP?

Imagine 1000 computers connected to the Internet.

Different computers may have:

- Different operating systems
- Different hardware
- Different network cards
- Different applications
- Different types of networks

They still need to communicate.

TCP/IP provides common rules that allow these different devices to communicate.

Example

Windows PC

|

|

TCP/IP

|

|

Linux Server

Both systems can communicate because they follow common networking protocols.

3. What is a Protocol?

A **protocol** is a set of rules used for communication.

Real-life example

When we talk to another person, we need a common language.

Person A → English → Person B

In networking:

Computer A → Protocol → Computer B

Examples:

- HTTP
- HTTPS
- TCP
- UDP
- IP

- DNS
 - DHCP
 - ARP
-

4. TCP/IP Model

The TCP/IP model is commonly explained using **4 layers**:

4. Application
3. Transport
2. Internet
1. Network Access

Easy way to remember

Application
Transport
Internet
Network Access

5. TCP/IP Layer 1 – Network Access

The Network Access layer is the lowest layer.

It deals with communication over the actual network.

It includes concepts related to:

- Ethernet
- Wi-Fi
- MAC addresses
- Network interface
- Physical transmission

Examples

Ethernet
Wi-Fi

Devices

Examples include:

- Network card
- Switch
- Wireless access point
- Cables

Data

Data is transmitted through the network using frames/signals.

6. TCP/IP Layer 2 – Internet Layer

The Internet layer is responsible for moving packets between networks.

The main protocol is:

IP – Internet Protocol

There are two major versions:

IPv4

IPv6

Example IPv4:

192.168.1.10

Example IPv6:

2001:db8::10

Main responsibilities

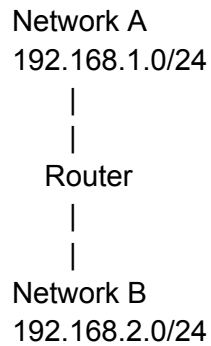
- IP addressing
- Packet delivery
- Routing
- Communication between different networks

Device

The main device associated with this layer is:

Router

Example:



The router forwards packets between networks.

7. TCP/IP Layer 3 – Transport Layer

The Transport layer provides communication between applications.

The two major protocols are:

TCP
UDP

TCP

TCP stands for:

Transmission Control Protocol

TCP provides reliable, connection-oriented communication.

It can:

- Establish a connection.
- Number data.
- Acknowledge received data.
- Retransmit missing data.
- Control the flow of data.

Example

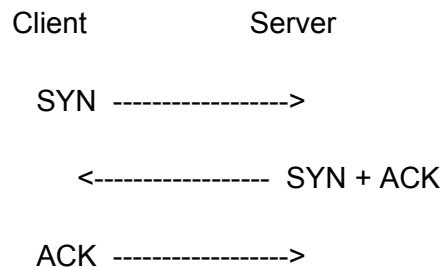
When downloading an important file, losing pieces of the file is a problem.

TCP helps provide reliable delivery.

8. TCP Three-Way Handshake

This is an important practical TCP concept.

Before TCP data communication starts, a connection is normally established using a three-way handshake.



Step 1 – SYN

Client says:

"I want to establish a TCP connection."

Step 2 – SYN + ACK

Server responds:

"I received your request and I am ready."

Step 3 – ACK

Client responds:

"Okay."

The TCP connection can now be used for communication.

9. UDP

UDP stands for:

User Datagram Protocol

UDP is connectionless and has less overhead than TCP.

It does not provide the same reliability mechanisms as TCP.

Examples

UDP is commonly used where low delay is important, such as:

- DNS queries
- Online gaming
- Voice communication
- Live streaming

Simple Difference

TCP → Reliable communication

UDP → Faster/low-overhead communication

Do not teach students that UDP is always "faster" in every situation. The important point is that UDP has less protocol overhead and does not provide TCP's reliability mechanisms.

10. Port Numbers

Transport protocols use **port numbers** to identify services/applications.

Example:

IP Address + Port

192.168.1.10:443

Some common ports:

Service	Protocol	Port
---------	----------	------

HTTP	TCP	80
HTTPS	TCP	443
DNS	UDP/TCP	53
SSH	TCP	22
FTP	TCP	21
DHCP	UDP	67/68

Real-life example

Think of an IP address as the **building address**.

The port is like the **specific room/service** inside that building.

IP Address → Which computer?

Port → Which application/service?

11. TCP/IP Layer 4 – Application Layer

The Application layer is the top layer.

It provides network services to applications.

Examples:

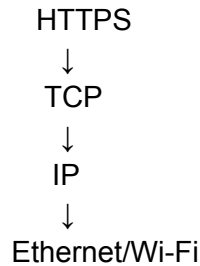
HTTP
HTTPS
DNS
FTP
SMTP
SSH
DHCP

Example: Web Browser

When you open a website:

Chrome / Edge / Firefox





This is a very important real-world example.

12. TCP/IP Model Complete Table

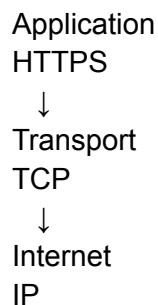
TCP/IP Layer	Main Function	Examples
Application	Network services for applications	HTTP, HTTPS, DNS, FTP, SSH
Transport	Application-to-application communication	TCP, UDP
Internet	IP addressing and routing	IPv4, IPv6, ICMP
Network Access	Local network and physical communication	Ethernet, Wi-Fi, ARP

13. TCP/IP Data Flow

Suppose a student opens:

`https://example.com`

The communication can be simplified as:



↓
Network Access
Ethernet/Wi-Fi

At the receiving computer, the process works in the reverse direction.

14. Encapsulation in TCP/IP

When data is sent, each layer adds information required for communication.

Simplified view:

Application Data
↓
TCP Header + Data
↓
IP Header + TCP Segment
↓
Ethernet Header + IP Packet
↓
Bits / Signals

For TCP:

Data
↓
TCP Segment
↓
IP Packet
↓
Ethernet Frame

This is called:

Encapsulation

At the receiving side:

Frame
↓
Packet
↓

Segment



Data

This is called:

Decapsulation

15. Important TCP Header Information

In Wireshark, when students open a TCP packet, they can see several fields.

Important fields to introduce:

Source Port

The port from which communication is coming.

Destination Port

The port where communication is going.

Sequence Number

Helps TCP keep track of data.

Acknowledgment Number

Helps TCP acknowledge received data.

Flags

Important TCP flags include:

SYN

ACK

FIN

RST

PSH

For Day 2, focus mainly on:

SYN
ACK
FIN
RST

16. Important IP Header Information

When students open the IP section in Wireshark, they can observe:

Source IP
Destination IP
TTL
Protocol

Example:

Source IP : 192.168.1.20
Destination IP : 8.8.8.8

Explain:

Source IP

Where the packet came from.

Destination IP

Where the packet is going.

TTL

Time To Live.

It helps prevent a packet from travelling through networks forever.

Protocol

Indicates the next protocol carried inside the IP packet, such as TCP, UDP, or ICMP.

17. Ethernet Header

Students can also open the Ethernet section.

They may see:

Destination MAC

Source MAC

Type

Example:

Source MAC : AA:BB:CC:11:22:33

Destination MAC : 11:22:33:44:55:66

This is related to the **Network Access layer** in the TCP/IP model.

Practical Part

Practical 1: Check TCP/IP Configuration

Open Command Prompt.

Run:

ipconfig

Ask students to identify:

IPv4 Address

Subnet Mask

Default Gateway

Example:

IPv4 Address : 192.168.1.20

Subnet Mask : 255.255.255.0

Gateway : 192.168.1.1

Explain:

IPv4 → Internet Layer
Network adapter → Network Access Layer

Practical 2: Check Network Adapter Information

Run:

```
ipconfig /all
```

Ask students to find:

Physical Address
IPv4 Address
Default Gateway
DNS Servers
DHCP Enabled

Explain:

- Physical Address → MAC address
 - IPv4 → IP address
 - Gateway → Router
 - DNS Server → Resolves domain names
-

Practical 3: Test IP Connectivity

Run:

```
ping 8.8.8.8
```

Students should observe:

Reply from 8.8.8.8

Explain that **ping** commonly uses **ICMP**, which works with the Internet layer.

Then run:

ping google.com

This also involves DNS before the ICMP communication can reach the resolved destination.

Practical 4: Check DNS

Run:

nslookup google.com

Students should observe the DNS server and returned address information.

Explain:

google.com

↓

DNS

↓

IP Address

DNS is an Application-layer protocol in the TCP/IP model.

Practical 5: Open Wireshark

Open Wireshark.

Select the active interface:

Wi-Fi

or

Ethernet

Start capturing.

Now generate some traffic.

Open a browser and visit:

<https://example.com>

After a few seconds, stop the capture.

Practical 6: Analyze a Packet

Select a packet.

Students will see something similar to:

Frame
Ethernet II
Internet Protocol Version 4
Transmission Control Protocol
TLS

This is the main practical concept for Day 2.

Ask students:

Which TCP/IP layer does each protocol belong to?

Example

TLS
↓
Application

TCP
↓
Transport

IPv4
↓
Internet

Ethernet
↓
Network Access

Practical 7: Filter TCP Packets

In Wireshark filter box:

tcp

Press Enter.

Now Wireshark shows TCP traffic.

Select one TCP packet.

Open:

Transmission Control Protocol

Find:

Source Port

Destination Port

Sequence Number

Acknowledgment Number

Flags

Practical 8: Observe TCP Three-Way Handshake

This is an excellent Day 2 practical.

Use this filter:

tcp.flags.syn == 1

You can also look for packets with:

SYN

SYN, ACK

ACK

Explain:

Client

Server

SYN ----->

<----- SYN + ACK

ACK ----->

Students should understand that this is how TCP establishes a connection.

Practical 9: Find HTTPS Traffic

Use:

```
tcp.port == 443
```

Students will see TCP traffic associated with HTTPS.

Explain:

443

↓

HTTPS

↓

Application

and:

TCP

↓

Transport

Practical 10: Analyze IP Header

Select an IP packet.

Open:

Internet Protocol Version 4

Find:

Source Address
Destination Address
TTL
Protocol

Students can write the values in their practical notebook.

Example:

Source IP : 192.168.1.20
Destination IP : 8.8.8.8
TTL : 117
Protocol : ICMP

Practical 11: Analyze Ethernet Header

Open:

Ethernet II

Find:

Destination
Source
Type

Explain:

Ethernet

↓

Network Access Layer

Students should understand that MAC addresses belong to local network communication.

Practical 12: Analyze UDP

Apply:

udp

Select a UDP packet.

Find:

Source Port
Destination Port
Length
Checksum

Then apply:

dns

If DNS traffic is present, students can inspect the DNS packet.

Explain:

DNS

↓

Application

UDP

↓

Transport

IP

↓

Internet

Ethernet

↓

Network Access

Practical 13: Analyze ICMP

Run:

ping 8.8.8.8

In Wireshark:

icmp

Students will see:

Echo Request
Echo Reply

Explain:

ICMP

↓

Internet Layer

Important:

ICMP is not TCP or UDP. It is a separate protocol carried by IP.

Practical 14: Follow a TCP Stream

This is a good beginner-intermediate Wireshark activity.

Steps

1. Capture traffic.
2. Find a TCP packet.
3. Right-click the packet.
4. Select **Follow** → **TCP Stream**.

Students can see the communication belonging to that TCP stream.

Explain that a TCP stream helps us study a particular TCP conversation.

OSI vs TCP/IP Model

Now connect Day 1 and Day 2.

OSI Model

OSI has:

7 Layers

Application

Presentation
Session
Transport
Network
Data Link
Physical

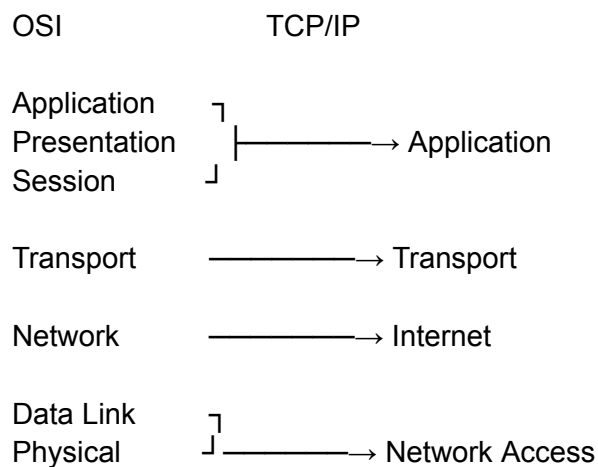
TCP/IP Model

TCP/IP is commonly represented using:

4 Layers
Application
Transport
Internet
Network Access

15. OSI vs TCP/IP Mapping

This is the most important comparison for Day 2.



16. Comparison Table

OSI Model	TCP/IP Model	Examples
Application	Application	HTTP, DNS, FTP

Presentation	Application	Encryption, data format
Session	Application	Session management
Transport	Transport	TCP, UDP
Network	Internet	IP, ICMP
Data Link	Network Access	Ethernet, MAC
Physical	Network Access	Cables, Wi-Fi signals

Important Point

The TCP/IP model combines some OSI layers.

OSI:
Application
Presentation
Session

TCP/IP:
Application

And:

OSI:
Data Link
Physical

TCP/IP:
Network Access

17. Real-World Example: Opening a Website

Suppose a student opens:

<https://www.example.com>

OSI View

Layer 7 → Application
Layer 6 → Presentation
Layer 5 → Session
Layer 4 → Transport
Layer 3 → Network
Layer 2 → Data Link
Layer 1 → Physical

TCP/IP View

Application
↓
Transport
↓
Internet
↓
Network Access

The actual communication is the same; the models organize and describe it differently.

18. Real-World Example: Sending a File

Suppose:

Computer A → Computer B

You send a file.

TCP/IP

Application
File transfer/application protocol
↓
Transport
TCP
↓
Internet
IP
↓
Network Access
Ethernet/Wi-Fi

OSI

Application
Presentation
Session
Transport
Network
Data Link
Physical

This demonstrates why the models are useful for understanding the same communication from different levels of detail.

19. Real-World Example: Online Gaming

When playing an online game:

Game Application



Transport
TCP/UDP



Internet
IP



Network Access
Wi-Fi/Ethernet

Many real-time applications use UDP because low delay can be important and the application can handle some aspects of reliability itself.

20. Real-World Example: DNS

Suppose the user enters:

google.com

The computer needs the IP address.

Application
DNS
↓
Transport
UDP/TCP
↓
Internet
IP
↓
Network Access
Ethernet/Wi-Fi

Students can actually see this communication in Wireshark.

21. Real-World Example: Video Call

During a video call:

Video Call Application
↓
Transport
UDP/TCP depending on the application
↓
Internet
IP
↓
Network Access
Wi-Fi / Ethernet

The exact protocols depend on the application.

22. Important Difference Between OSI and TCP/IP

Tell students:

OSI

- 7 layers
- Mainly a reference/modeling framework
- Separates Presentation and Session as separate layers
- Separates Physical and Data Link

TCP/IP

- Commonly represented using 4 layers
 - Based around protocols used in real Internet networking
 - Combines Application, Presentation and Session into Application
 - Combines Data Link and Physical into Network Access
-

23. Important Interview Questions

Q1. What is TCP/IP?

TCP/IP is a suite of networking protocols used for communication between devices.

Q2. How many layers are commonly used in the TCP/IP model?

Four.

Application

Transport

Internet

Network Access

Q3. Which protocol is used for reliable transport?

TCP.

Q4. Which protocol is connectionless?

UDP.

Q5. Which protocol is used for IP addressing?

IP.

Q6. Which device commonly forwards packets between networks?

Router.

Q7. Which protocol resolves domain names?

DNS.

Q8. What is the default HTTPS port?

443.

Q9. What is the difference between IP and MAC?

IP → Network communication/routing

MAC → Local network communication

Q10. What is encapsulation?

Adding protocol information as data moves down the networking layers before transmission.

Day 2 Final Practical Task

Give students this complete activity.

Task: Analyze TCP/IP Traffic Using Wireshark

Part 1 – Network Information

Run:

```
ipconfig /all
```

Record:

- IPv4 address
- MAC address
- Default gateway
- DNS server

Part 2 – Test Network

Run:

ping 8.8.8.8

Observe the result.

Part 3 – DNS

Run:

nslookup google.com

Observe the response.

Part 4 – Capture Traffic

Open Wireshark and start capturing.

Open:

<https://example.com>

Stop the capture.

Part 5 – TCP Analysis

Apply:

tcp

Find:

- Source port
- Destination port
- TCP flags
- Sequence number
- Acknowledgment number

Part 6 – IP Analysis

Open an IPv4 packet and record:

- Source IP
- Destination IP
- TTL
- Protocol

Part 7 – Ethernet Analysis

Record:

- Source MAC
- Destination MAC
- Ethernet type

Part 8 – DNS Analysis

Apply:

dns

Find a DNS packet and identify:

- Query name
- Source
- Destination

Part 9 – TCP/IP Layer Mapping

Complete:

Protocol TCP/IP Layer

HTTP/HTTPS	_____
TCP	_____
UDP	_____
IPv4	_____
Ethernet	_____

Day 2 Student Assignment

Compare OSI and TCP/IP

Students should create a table:

OSI Layer	TCP/IP Layer	Example
Application	Application	HTTP
Presentation	Application	Encryption

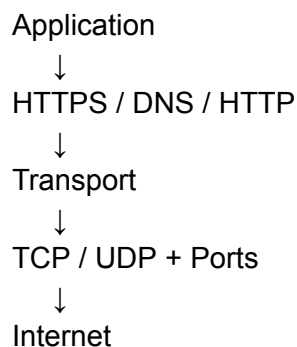
Session	Application	Session
Transport	Transport	TCP
Network	Internet	IP
Data Link	Network Access	Ethernet
Physical	Network Access	Cable/Wi-Fi

Then answer:

1. What is TCP/IP?
2. What is a protocol?
3. What are the four TCP/IP layers?
4. What is TCP?
5. What is UDP?
6. What is an IP address?
7. What is a port?
8. What is TCP three-way handshake?
9. What is encapsulation?
10. How are OSI and TCP/IP models different?
11. Which TCP/IP layer contains TCP?
12. Which TCP/IP layer contains IP?
13. Which TCP/IP layer contains Ethernet?
14. Which protocol is used for DNS?
15. What information can be observed in a TCP packet using Wireshark?

Final Concept Students Should Understand

By the end of Day 2, students should be able to look at a real Wireshark packet and mentally understand:



↓
IP + Routing
↓
Network Access
↓
Ethernet / Wi-Fi + MAC

That gives them a strong practical connection between **Day 1 OSI theory** and **Day 2 real TCP/IP traffic analysis**.